

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.¹ In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$$H_0: p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = p_7 = \frac{1}{7}$$

$$H_a: \text{at least one } p_i \neq \frac{1}{7}, i = 1 \sim 7$$

$$\chi^2_{\text{STAT}} = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi^2_{0.05, 6}$$

$$E_i = \frac{200}{7} = 28.57$$

$$\chi^2_x = \frac{(24 - 28.57)^2}{28.57} + \frac{(36 - 28.57)^2}{28.57} + \dots + \frac{(29 - 28.57)^2}{28.57} = 3.63$$

$$\chi^2_{0.05, 6} = 12.5916$$

$$RR: \{ \chi^2 \mid \chi^2 > 12.5916 \}$$

$$12.5916 > 3.63 \Rightarrow \text{non-reject } H_0$$

There is no sufficient evidence to say that heart attack incidence differs by day